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ON ALEKSANDROV TYPE PROBLEM FOR UNBOUNDED c-CONVEX POLYHEDRA

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Abstract. For a function $c : \mathbb{S}^n \times \mathbb{S}^n \to \mathbb{R} \cup \{\infty\}$ with proper structure, we study unbounded c-convex polyhedra in $\mathbb{R}^{n+1}$ as the envelope of a family of c-hyperplanes. Such objects arise naturally in the geometry of hypersurfaces and the theory of optimal transportation. In this paper, we show the existence of unbounded c-convex polyhedra with prescribed c-curvature and asymptotic cone.

1. **Introduction.** In Chapter 9 of [1] Aleksandrov studied a problem of unbounded hypersurfaces with prescribed Gauss curvature. By choosing the coordinate, a complete and non-compact convex hypersurface $\mathcal{M}$ can be regarded as a complete graph of a convex function $u : \Omega \to \mathbb{R}$, where Ω is a convex set of $\mathbb{R}^n$. For each Borel $\omega \subset \Omega$, Aleksandrov introduced

$$\tilde{\mu}_{\mathcal{M}}(\omega) := \sigma_{\mathbb{S}^n}\big(\{\nu_{\mathcal{M}}\big(z, u(z)\big) : z \in \omega\}\big), \tag{1}$$

where $\nu_{\mathcal{M}}((z, u(z)))$ is the set of unit outer normals of $\mathcal{M}$ at $(z, u(z))$ and $\sigma_{\mathbb{S}^n}$ is the standard volume measure of $\mathbb{S}^n$. Suppose that the origin $\mathcal{O}$ lies in the interior of $\mathcal{D}_{\mathcal{M}} = \big\{(z,t) \in \mathbb{R}^{n+1} : z \in \Omega, t \geq u(z)\big\}$. The collection of all rays starting from $\mathcal{O}$ and contained in $\mathcal{D}_{\mathcal{M}}$ forms a solid convex cone in $\mathbb{R}^{n+1}$, which is called the asymptotic cone of $\mathcal{M}$. Aleksandrov studied the following question: given a Borel measure γ on $\mathbb{R}^n$ and a convex cone $\mathcal{C}$ in $\mathbb{R}^{n+1}$, is there a complete and non-compact convex hypersurface $\mathcal{M}$ in $\mathbb{R}^{n+1}$ with asymptotic cone $\mathcal{C}$ such that $\tilde{\mu}_{\mathcal{M}} = \gamma$? Aleksandrov [1] obtained the sharp conditions for the existence of solutions to his problem. Following [11, 12], one may formulate the problem as a PDE of the graph function u. Recall that the volume form and Gaussian curvature of $\mathcal{M}$ at $(x, u(x))$ are respectively

$$d\,\mathrm{vol}_{\mathcal{M}}(z) = (1 + |Du|^2)^{\frac{1}{2}} dz \quad \text{and} \quad K_{\mathcal{M}}(z) = \det D^2 u/(1 + |Du|^2)^{\frac{n}{2}+1}.$$

Then $\tilde{\mu}_{\mathcal{M}} = f dz$ is equivalent to the following Monge-Ampère equation

$$\det D^2 u(z) = \frac{f(z)}{g(Du(z))} \quad \text{on } \Omega, \text{ where } g(z') := (1 + |z'|^2)^{-\frac{n+1}{2}}. \tag{2}$$

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Since $\nu_{\mathcal{M}}(x) = (Du, -1)/\sqrt{1+|Du|^2}$, one sees that

$$Du(\Omega) = \mathcal{C}^* \cap \{(z,t) \in \mathbb{R}^{n+1} : t = -1\} =: \Omega^*,$$

where $\mathcal{C}^* := \{q \in \mathbb{R}^{n+1} : \langle p, q\rangle \leq 0, \forall p \in \mathcal{C}\}$ is the polar cone of $\mathcal{C}$. Then the problem is an optimal transportation from $(\Omega, f(z)dz)$ to $(\Omega^*, g(z')dz')$ with cost function $c(z,z') = z \cdot z'$ [11, 12]. The regularity of the solution was studied in [6, 14].

In this paper, we are interested in a problem analogous to the above one. Fix a convex cone $\mathcal{C}$ of $\mathbb{R}^{n+1}$. Let $\mathcal{M}$ be a complete, non-compact convex hypersurface with $\mathcal{D}_{\mathcal{M}}$ containing the origin $\mathcal{O}$ in its interior, and $\mathcal{C}$ being its asymptotic cone. Then $\mathcal{M}$ can be parametrized by its radial graph $\vec{\varrho}(x) = \varrho(x)x$ over $x \in \Omega$, where $\vec{\varrho}(x) = \mathcal{M} \cap \{tx : t \geq 0\}$ and $\Omega = \mathbb{S}^n \setminus \overline{\mathcal{C}}$. Aleksandrov [1, 2] defined his integral Gaussian curvature as

$$\mu_{\mathcal{M}}(\omega) := \sigma_{\mathbb{S}^n}(\{\nu_{\mathcal{M}}(\vec{\varrho}(x)) : x \in \omega \cap \Omega\}), \text{ where } \omega \text{ is any Borel subset of } \mathbb{S}^n. \quad (3)$$

We are interested in the following problem: given a Borel measure γ on Ω, if there is a convex hypersurface $\mathcal{M}$ in $\mathbb{R}^{n+1}$ with asymptotic cone $\mathcal{C}$ such that

$$\mu_{\mathcal{M}} = \gamma \quad \text{on } \Omega. \quad (4)$$

When $\Omega = \Omega^* = \mathbb{S}^n$, the problem of finding a closed hypersurface with the prescribed integral Gaussian curvature (4) has been resolved by Aleksandrov [2], and a variational approach was later on found in [13] as it is an optimal transportation on $\mathbb{S}^n$ with cost $c(x,y) = \log\cos d_{\mathbb{S}^n}(x,y)$. Recently it has also been generalized to the so-called Gauss image problem by Böröczky et al. [3]. While for the case of complete and non-compact hypersurfaces, very little is known. Suppose $\mathcal{M}$ is a complete graph of a convex function u on $\mathbb{R}^n$. Let $X(z) = (z, u(z)) \in \mathcal{M}$. Then

$$|X| = (|z|^2 + u^2)^{\frac{1}{2}} \quad \text{and} \quad \langle X, \nu\rangle = \frac{z \cdot Du - u}{\sqrt{1+|Du|^2}}.$$

According to [4, 8], $\mu_{\mathcal{M}} = f dx$ is equivalent to the following Monge-Ampère equation

$$\det D^2 u = \frac{(1+|Du|^2)^{\frac{n+1}{2}}(z\cdot Du - u)}{(|z|^2+|Du|^2)^{\frac{n+1}{2}}} f\Big(\frac{(z,u)}{\sqrt{|z|^2+u^2}}\Big) \quad \text{in } \mathbb{R}^n. \quad (5)$$

It is straightforward to see from (2) and (5) that the two problems are different.

Let us consider a class of c-convex hypersurfaces with prescribed c-curvature measure, which includes the problem (4) as a special case. Given $d_0 \in (0,\pi)$, let $\mathscr{C}_{d_0}$ be the class of functions on $\mathbb{S}^n \times \mathbb{S}^n$ in the form

$$c(x,y) = \begin{cases} F(d_{\mathbb{S}^n}(x,y)), & \text{if } d_{\mathbb{S}^n}(x,y) < d_0, \\ -\infty, & \text{if } d_{\mathbb{S}^n}(x,y) \geq d_0, \end{cases} \quad (6)$$

where $F \in C^1([0,d_0))$ is a strictly concave function satisfying

$$\lim_{t \nearrow d_0} F(t) = -\infty, \qquad \int_0^{d_0} F(t)dt > -\infty. \quad (7)$$

We recall some notions as in [9]. Consider the function $\Gamma : \mathbb{S}^n \times \mathbb{R}^{n+1} \to \mathbb{R}$ given by

$$\Gamma(y,p) := |p| e^{c(p/|p|, y)}. \quad (8)$$

It is well-defined at $p = \mathcal{O}$, since (7) implies that $c(x, y)$ is bounded from above and thus $\lim_{|p|\to 0} \Gamma(y, p) = 0$. Fixing $y \in \mathbb{S}^n$ and $a > 0$, the following hypersurface and region are called *c-hyperplane* and *c-subspace* respectively

$$\begin{aligned} \ell_{y,a} &= \{p \in \mathbb{R}^{n+1} : \Gamma(y, p) = a\}, \\ \mathcal{H}_{y,a} &= \{p \in \mathbb{R}^{n+1} : \Gamma(y, p) \leq a\}. \end{aligned} \tag{9}$$

A region $\mathcal{D}$ of $\mathbb{R}^{n+1}$ is called a *c-convex body* if

$$\mathcal{D} = \cap_{y\in\mathcal{I}} \mathcal{H}_{y,a_y}, \tag{10}$$

where $\mathcal{I} \subset \mathbb{S}^n$ is an index set and $a_y > 0$. We call $\mathcal{M}$ a *c-convex hypersurface* if $\mathcal{M} = \partial\mathcal{D}$ with $\mathcal{D}$ being a c-convex body. Since $\Gamma(y, tp)$ is of homogeneous degree one in t, we see that $\ell_{y,a} \cap \{tx : x \in \mathbb{S}^n, t \geq 0\}$ is either empty or a single point. This together with $\mathcal{O} \in \mathcal{D}$ implies that every c-convex hypersurface $\mathcal{M}$ is star-shaped w.r.t. $\mathcal{O}$. Let $\Omega = \{x \in \mathbb{S}^n : \exists t > 0 \ s.t. \ tx \in \mathcal{M}\}$. Then there is a function $r_{\mathcal{M}} : \Omega \to (0, \infty)$ such that

$$\mathcal{M} = \big\{r_{\mathcal{M}}(x)x : x \in \Omega\big\}.$$

Then c-Gauss map $\mathscr{A}^c_{\mathcal{M}} : \Omega \to \mathbb{S}^n$ of a c-convex hypersurface $\mathcal{M}$ is given by

$$\mathscr{A}^c_{\mathcal{M}}(x) = \big\{y \in \mathbb{S}^n : \ \Gamma(y, p) \leq \Gamma\big(y, r_{\mathcal{M}}(x)x\big) \ \forall p \in \mathcal{M}\big\}. \tag{11}$$

Definition 1.1. Let us introduce some notions.

(i) A subset ω of $\mathbb{S}^n$ is called d_0-*convex* if there exists an index set $\mathcal{I} \subset \mathbb{S}^n$ such that $\omega = \cap_{x\in\mathcal{I}} V_{x,d_0}$, where $V_{x,d_0} := \{y \in \mathbb{S}^n : d_{\mathbb{S}^n}(x, y) \geq d_0\}$.

(ii) For $\omega \subseteq \mathbb{S}^n$, its d_0-*dual* is given by $\omega^\sharp := \cap_{y\in\omega} V_{y,d_0}$.

(iii) A cone $\mathcal{C}$ with vertex $\mathcal{O}$ is called a d_0-*convex* cone if $\mathcal{C} \cap \mathbb{S}^n$ is d_0-convex. If, in addition, the index set is finite, $\mathcal{C}$ is a d_0-*polyhedral angle*.

Fix a d_0-convex cone $\mathcal{C}$. Take

$$\Omega = \Omega(\mathcal{C}) = \mathbb{S}^n \setminus \mathcal{C} \quad \text{and} \quad \Omega^* = \Omega^*(\mathcal{C}) = (\mathcal{C} \cap \mathbb{S}^n)^\sharp, \tag{12}$$

and denote $\mathscr{M}(\mathcal{C}) = \big\{c\text{-convex hypersurface } \mathcal{M} = \partial\mathcal{D} \text{ with } \mathcal{I} = \Omega^* \text{ in (10)}\big\}$. By (7) and (8), $\ell_{y,a}$ is unbounded and lies outside the cone $\cup_{t\geq 0}\big[tV_{y,d_0}\big]$; and for any sequence of points $p_i \in \ell_{y,a}$, if $|p_i| \to \infty$ and $\lim_{i\to\infty} p_i/|p_i| = x$ then $d_{\mathbb{S}^n}(x, y) = d_0$. It follows that for any $\mathcal{M} \in \mathscr{M}(\mathcal{C})$, if $\mathcal{D}$ is the c-convex body whose boundary is $\mathcal{M}$, then $\mathcal{D} \supseteq \mathcal{C}$. Such a cone $\mathcal{C}$ is thus called the *asymptotic cone* of c-convex hypersurfaces in $\mathscr{M}(\mathcal{C})$. For each $\mathcal{M} \in \mathscr{M}(\mathcal{C})$, associated is the following c-curvature measure

$$\mu^c_{\mathcal{M}}(\omega) = \sigma_{\mathbb{S}^n}\big(\mathscr{A}^c_{\mathcal{M}}(\omega \cap \Omega)\big), \text{ for any Borel set } \ \omega \subset \mathbb{S}^n. \tag{13}$$

It is not hard to see that $\mu^c_{\mathcal{M}}$ is a measure on Ω [9]. Given a finite Borel measure γ supported on Ω, we are interested in whether there is a $\mathcal{M} \in \mathscr{M}(\mathcal{C})$ such that

$$\mu^c_{\mathcal{M}} = \gamma. \tag{14}$$

When $c = \log\cos d_{\mathbb{S}^n}$, problem (14) reduces to (4).

In this paper, we solve (14) in the discrete case

$$\gamma = \sum_{i=1}^N \gamma_i \delta_{x_i}, \text{ where } X = \{x_i\}_{i=1}^N \subseteq \mathbb{S}^n. \tag{15}$$

Let $\mathcal{C}$ be a d_0-polyhedral angle, and Ω, Ω^* be as in (12). Suppose

$$X \subset \Omega. \tag{16}$$

Given $\mathcal{P} = \{p_i = t_i x_i : t_i > 0, i = 1, \dots, N\}$, its *c-convex hull* with respect to $\mathcal{C}$ is defined as $\mathbb{K}_{\mathcal{P}} = \partial(\langle \mathcal{P} \rangle_{\mathcal{C}})$, where

$$\langle \mathcal{P} \rangle_{\mathcal{C}} = \bigcap_{y \in \bar{\Omega}^*} \Big\{ p \in \mathbb{R}^{n+1} : \Gamma(y, p) \le \max_{1 \le i \le N} \Gamma(y, p_i) \Big\}. \tag{17}$$

Clearly $\mathbb{K}_{\mathcal{P}} \in \mathscr{M}(\mathcal{C})$, and we call it a *c-convex polyhedron with vertices $\mathcal{P}$ and limit angle $\mathcal{C}$*. The geometric meaning of the limit angle is explained in Lemma 2.1. Suppose

$$X \cap B_{d_0}(y) \neq \emptyset, \ \forall y \in \bar{\Omega}^*, \tag{18}$$

which is compatible with (i) and (ii). This implies that for all $y \in \bar{\Omega}^*$, there is some $p_i \in \mathcal{P}$ such that $\Gamma(y, p_i) = t_i e^{c(x_i, y)} > 0$. Hence $\mathcal{O}$ lies in the interior of $\langle \mathcal{P} \rangle_{\mathcal{C}}$. Let

$$\mathscr{K}(X, \mathcal{C}) = \big\{ \mathbb{K}_{\mathcal{P}} : \ \mathcal{P} = \{p_i = t_i x_i : t_i > 0, i = 1, \dots, N\} \big\}.$$

One will see in Lemma 2.3 that c-curvature measures of c-convex polyhedra in $\mathscr{K}(X, \mathcal{C})$ must be linear combinations of Dirac measures at x_i. Our main theorem provides a complete solution to problem (14) with γ being (15).

Theorem 1.2. *Suppose $c \in \mathscr{C}_{d_0}$. Let $\mathcal{C}$ be a d_0-polyhedral angle, Ω, Ω^* be as in (12), and γ be a measure of the form (15). Suppose (16) and (18) hold. Then there exists $\mathbb{K} \in \mathscr{K}(X, \mathcal{C})$ such that its c-curvature measure $\mu^c_{\mathbb{K}} = \gamma$ if and only if*

(i) *$\gamma(\Omega) = \sigma_{\mathbb{S}^n}(\Omega^*)$;*
(ii) *$\gamma(\omega) < \sigma_{\mathbb{S}^n}(\omega_{d_0} \cap \Omega^*)$ for any d_0-convex $\omega \subset \mathbb{S}^n$, where $\omega_{d_0} := \{x \in \mathbb{S}^n : d_{\mathbb{S}^n}(x, \omega) < d_0\}$ is the d_0-neighborhood of ω in $\mathbb{S}^n$.*

Remark 1.3. We show that conditions (i) and (ii) are necessary. Suppose $\mathbb{K} \in \mathscr{K}(X, \mathcal{C})$ satisfies $\mu^c_{\mathbb{K}} = \gamma$. By (11), $\mathscr{A}^c_{\mathbb{K}}(\Omega) = \Omega^*$. It follows by (13) that

$$\gamma(\Omega) = \mu^c_{\mathbb{K}}(\Omega) = \sigma_{\mathbb{S}^n}(\mathscr{A}^c_{\mathbb{K}}(\Omega)) = \sigma_{\mathbb{S}^n}(\Omega^*).$$

This gives necessity of condition (i). Using (7) and (11), we have $d_{\mathbb{S}^n}(x, \mathscr{A}^c_{\mathbb{K}}(x)) < d_0$ for all $x \in \Omega$. Hence $\mathscr{A}^c_{\mathbb{K}}(\omega \cap \Omega)$ is strictly contained in $\omega_{d_0} \cap \Omega^*$. Therefore

$$\gamma(\omega) = \mu^c_{\mathbb{K}}(\omega) = \sigma_{\mathbb{S}^n}(\mathscr{A}^c_{\mathbb{K}}(\omega \cap \Omega)) < \sigma_{\mathbb{S}^n}(\omega_{d_0} \cap \Omega^*).$$

The problem of closed c-convex hypersurface with prescribed c-curvature was first studied in [9]. In the current paper, we focus on unbounded c-convex hypersurfaces. In [10], the authors studied the case that Ω and Ω^* are subdomains of $\mathbb{S}^n$ satisfying the position condition

$$\Omega \cap B_{d_0}(y) \neq \emptyset \quad \text{and} \quad \Omega^* \cap B_{d_0}(x) \neq \emptyset, \quad \forall (x, y) \in \bar{\Omega} \times \bar{\Omega}^*. \tag{19}$$

This condition is very natural as it means that the c-Gauss map (11) is defined on $\bar{\Omega}$. In this paper, we assume Ω and Ω^* satisfy (12). This condition can be viewed as the *extremal* case of (19), as it implies that (19) holds for all $(x, y) \in \Omega \times \Omega^*$ but fails only on the boundaries of Ω, Ω^*. Our Theorem 1.2 addresses such *extremal* case.

Besides the aforementioned problem (4), another significant example covered by Theorem 1.2 is the problem of light refraction. Given illumination density $f(x)$ on $\Omega \subset \mathbb{S}^n$, one seeks a refractor in $\mathbb{R}^{n+1}$ separating two media with refractive index κ, such that the refracted rays emerging in $\Omega \subset \mathbb{S}^n$ have prescribed density $g(y)$. When $\kappa > 1$, Gutiérrez and Huang [7] reformulated this as an optimal transportation from (Ω, f) to (Ω^*, g) with cost $c(x, y) = \log(\kappa \cos d_{\mathbb{S}^n}(x, y) - 1) \in \mathscr{C}_{\arccos(1/\kappa)}$. Hence Theorem 1.2 would have some application: given a set of atomic illuminations, one may design an unbounded polyhedral refractor such that the light beams are uniformly refracted into a prescribed region. The existence result in [7] needs the domains Ω and Ω^* satisfy

$$\inf_{(x,y)\in\Omega\times\Omega^*} x\cdot y \geq 1/\kappa+\delta.$$

This constraint does not hold in our situation (12), and so Theorem 1.2 also provides a new existence result for refractors.

Our proof for Theorem 1.2 is variational. We formulate the prescribing c-curvature measure problem as an optimal transport problem and analyze the existence of potentials for the associated Kantorovich functional. The main difficulty lies in the absence of global C^0 estimates for the potential functions. This will be overcome by exploiting the discrete structure of unbounded c-convex polyhedra. Although global bound of the radial potential is unattainable, the distances from the vertices to the origin $\mathcal{O}$ can be effectively controlled, which is sufficient to establish the existence of the desired c-convex polyhedra. This crucial discrete oscillation estimate is proved in Lemma 3.3.

This paper is organized as follows. In Section 2, we recall some basic properties of d_0-convex sets, unbounded c-convex polyhedra, c-support functions and c-curvature measures. Section 3 is devoted to oscillation estimates for potential functions, based on a strengthened Aleksandrov condition and Kantorovich duality. Section 4 gives the proof of Theorem 1.2.

2. **Preliminaries.** This section gives some settings for Theorem 1.2. We first explain the geometric nature of asymptotic cones.

Lemma 2.1. *Let $\mathcal{C}$ be the limit angle of $\mathbb{K}\in\mathscr{K}(X,\mathcal{C})$. Then $\lim_{\lambda\searrow 0}\lambda\mathbb{K}=\mathcal{C}$ in the locally Hausdorff sense.*

Proof. Since $\Gamma(y,\lambda p)=\lambda\Gamma(y,p)$ for any $\lambda>0$, we have $\ell_{x,\lambda a}=\lambda\ell_{x,a}$, where the function Γ and the hypersurface $\ell_{y,a}$ are defined in (8) and (9), respectively. Let $\mathbf{B}_R$ denote the Euclidean ball in $\mathbb{R}^{n+1}$ with center $\mathcal{O}$ and radius R. Then $p\in\ell_{y,\lambda a}\cap\mathbf{B}_R$ if and only if $\Gamma(y,p)=\lambda a$ and $|p|=R$, which is equivalent to $e^{c(y,p/R)}=\lambda a/R$. Fixing $R>0$ and then passing $\lambda\searrow 0$, it follows from (6) and (7) that $d_{\mathbb{S}^n}(y,p/R)\nearrow d_0$. Hence

$$\lim_{\lambda\searrow 0}(\lambda\ell_{y,a}\cap\mathbf{B}_R)=R\cdot V_{y,d_0},$$

where V_{y,d_0} is in Definition 1.1. We further obtain

$$\lim_{\lambda\searrow 0}\lambda\cdot\{p\in\mathbb{R}^{n+1}:\Gamma(y,p)\leq a\}=\Sigma_y,$$

where $\Sigma_y:=\{tx: t\geq 0, x\in V_{y,d_0}\}$. Let $p_1,\dots,p_N$ be the vertices of $\mathbb{K}$. Using (17),

$$\lim_{\lambda\searrow 0}\lambda\mathbb{K}=\bigcap_{y\in\bar{\Omega}^*}\lim_{\lambda\searrow 0}\lambda\cdot\left\{p\in\mathbb{R}^{n+1}:\Gamma(y,p)\leq\max_{1\leq i\leq N}\Gamma(y,p_i)\right\}=\bigcap_{y\in\bar{\Omega}^*}\Sigma_y=\mathcal{C}.$$

$\square$

Since $\ell_{y,a}\cap\{tx: x\in\mathbb{S}^n, t\geq 0\}$ is either empty or a single point, we see from (17) that $\mathbb{K}\in\mathscr{K}(X,\mathcal{C})$ is star-shaped. Its *radial function* $r_{\mathbb{K}}$ is then defined by $r_{\mathbb{K}}(x):=\sup\{\lambda>0:\lambda x\in\mathbb{K}\}$ for each $x\in\Omega$. The associated *c-support function* is

$$u_{\mathbb{K}}(y):=\sup_{x\in\Omega}\Gamma(y,r_{\mathbb{K}}(x)x),\quad\forall y\in\bar{\Omega}^*. \tag{20}$$

In particular, when $c(x,y) = \log\cos d_{\mathbb{S}^n}(x,y)$, the function $u_{\mathbb{K}}$ is the support function of the convex polyhedron $\mathbb{K}$. The following lemma says that any member of $\mathbb{R}^N \times C(\bar{\Omega}^*)$ determines an unbounded c-convex polyhedron in $\mathscr{K}(X,\mathcal{C})$. This characterization originates from convex geometry and optimal transport theory

Lemma 2.2. *Suppose $(\varrho,\psi) \in \mathbb{R}^N \times C(\bar{\Omega}^*)$ satisfies*

$$\varrho^i = \max_{y\in\bar{\Omega}^*}\{c(x_i,y) - \psi(y)\}, \quad 1 \le i \le N, \tag{21}$$

$$\psi(y) = \max_{1\le i\le N}\{c(x_i,y) - \varrho^i\}, \quad \forall y \in \bar{\Omega}^*, \tag{22}$$

where ϱ^i is the i-th component of $\varrho \in \mathbb{R}^N$. Then the c-convex hull of $\mathcal{P} = \{e^{-\varrho^i}x_i\}_{i=1}^N$ with respect to $\mathcal{C}$, denoted by $\mathbb{K}$, is an unbounded c-convex polyhedron in $\mathscr{K}(X,\mathcal{C})$ such that $\varrho^i = -\log r_{\mathbb{K}}(x_i)$ and $\psi = \log u_{\mathbb{K}}$, where $r_{\mathbb{K}}$ and $u_{\mathbb{K}}$ are the radial function and the c-support function of $\mathbb{K}$, respectively. In this case, $\mathbb{K}$ is called the unbounded c-convex polyhedron associated to the pair (ϱ,ψ).

Proof. Suppose $\mathbb{K} \in \mathscr{K}(X,\mathcal{C})$ has vertices $p_i := e^{-\varrho^i}x_i$. It follows from (17) that

$$\mathbb{K} = \langle\{p_i\}_{i=1}^N\rangle_{\mathcal{C}} = \bigcap_{y\in\bar{\Omega}^*}\left\{p \in \mathbb{R}^{n+1} : \Gamma(y,p) \le \max_{1\le i\le N}\Gamma(y,p_i)\right\}. \tag{23}$$

On the other hand, let $u_{\mathbb{K}}$ be the c-support function of $\mathbb{K}$ defined as in (20). Then

$$\mathbb{K} = \bigcap_{y\in\bar{\Omega}^*}\left\{p \in \mathbb{R}^{n+1} : \Gamma(y,p) \le u_{\mathbb{K}}(y)\right\}. \tag{24}$$

Comparing (23) and (24), we conclude that $u_{\mathbb{K}}(y) = \max_{1\le i\le N}\Gamma(y,p_i)$. Recalling the definition of Γ (8) and combining (21), we further see that $\psi = \log u_{\mathbb{K}}$.

Note that $\Gamma(y,\lambda p) = \lambda\Gamma(y,p)$ for any $p \in \mathbb{R}^{n+1}$ and any $\lambda > 0$. Then the relation in the right-hand side of (23) becomes $1/|p| \ge \frac{\Gamma(y,p/|p|)}{\max_{1\le i\le N}\Gamma(y,p_i)}$, which implies that

$$\frac{1}{r_{\mathbb{K}}(x)} = \max_{y\in\bar{\Omega}^*}\frac{\Gamma(y,x)}{\max_{1\le i\le N}\Gamma(y,p_i)}.$$

In particular, taking $x = x_i$, we have $\varrho^i = -\log r_{\mathbb{K}}(x_i)$ for each $1 \le i \le N$. □

By virtue of Lemma 2.2, we can show that the c-curvature measure of a c-convex polyhedron is the linear combination of Dirac measures. For simplicity of notation, we will use $|\cdot|$ to denote the standard Riemannian measure $\sigma_{\mathbb{S}^n}$ on $\mathbb{S}^n$.

Lemma 2.3. *Given $\mathbb{K} \in \mathscr{K}(X,\mathcal{C})$, let $\mu^c_{\mathbb{K},i} := |\mathscr{A}^c_{\mathbb{K}}(x_i)|$ for each $1 \le i \le N$. Then the c-curvature measure $\mu^c_{\mathbb{K}}$ is given by $\sum_{i=1}^N \mu^c_{\mathbb{K},i}\delta_{x_i}$.*

Proof. Define $\mathcal{Y}_{\mathbb{K}}$ to be the set of points $y \in \bar{\Omega}^*$ such that there are two distinct points $x_1, x_2 \in \Omega$ satisfying $y \in \mathscr{A}^c_{\mathbb{K}}(x_1) \cap \mathscr{A}^c_{\mathbb{K}}(x_2)$. Then the support function u of $\mathbb{K}$ is not differentiable at y. By semi-convexity of u, the singular set of u is negligible. In particular, $\sigma_{\mathbb{S}^n}(\mathcal{Y}_{\mathbb{K}}) = 0$. Let $B_\delta(x) \subset \Omega$ be a geodesic ball with $B_\delta(x) \cap \{x_i\}_{i=1}^N = \emptyset$. Given any $x' \in B_\delta(x)$ and any $y \in \mathscr{A}^c_{\mathbb{K}}(x')$, using (22) there exists at least one x_i such that $y \in \mathscr{A}^c_{\mathbb{K}}(x_i)$. Then $\mathscr{A}^c_{\mathbb{K}}(B_\delta(x)) \subset \mathcal{Y}_{\mathbb{K}}$. Thus $\mu^c_{\mathbb{K}}(B_\delta(x)) = \sigma_{\mathbb{S}^n}(\mathscr{A}^c_{\mathbb{K}}(B_\delta(x))) = 0$. Since $\delta > 0$ can be arbitrarily small, we conclude that $\mu^c_{\mathbb{K}} = \sum_{i=1}^N \mu^c_{\mathbb{K},i}\delta_{x_i}$. □

Let $\mathcal{A}$ be the set of $(\varrho, \psi) \in \mathbb{R}^N \times C(\bar{\Omega}^*)$ satisfying (21) and (22). Consider the functional

$$\mathcal{J}(\varrho, \psi) := \sum_{i=1}^{N} \gamma_i \varrho^i + \int_{\Omega^*} \psi(y) d\sigma_{\mathbb{S}^n}(y), \quad \forall (\varrho, \psi) \in \mathcal{A}. \tag{25}$$

Here $d\sigma_{\mathbb{S}^n}$ denotes the n-dimensional Hausdorff measure on $\mathbb{S}^n$. Note that $\mathcal{A}$ is non-empty because an arbitrary unbounded convex polyhedron $\mathbb{K} \in \mathscr{K}(X, \mathcal{C})$ induces a member $(\varrho, \psi) \in \mathcal{A}$ by setting $\varrho^i := -\log r_{\mathbb{K}}(x_i)$ and $\psi := \log u_{\mathbb{K}}$, where $r_{\mathbb{K}}$ and $u_{\mathbb{K}}$ are the radial function and the c-support function of $\mathbb{K}$, respectively. The mass balance condition (i) in Theorem 1.2 indicates that the functional $\mathcal{J}$ is invariant under additive constants, namely $\mathcal{J}(\varrho, \psi) = \mathcal{J}(\varrho - a, \psi + a)$. It is shown in section 4 that minimizers of $\mathcal{J}$ in $\mathcal{A}$ induce solutions to the Aleksandrov type problem (14). The existence of minimizers is a consequence of oscillation estimates in Lemma 3.3.

We turn to show some basic properties of c-convex functions and c-curvature measures, which basically follow from [9]. The following lemma reduces gradient estimates for c-convex functions to oscillation estimates.

Lemma 2.4. *Let $c \in \mathscr{C}_{d_0}$. Given $(\varrho, \psi) \in \mathcal{A}$, let*

$$d_\psi := \sup\{d_{\mathbb{S}^n}(x_i, y) : y \in \bar{\Omega}^*, \psi(y) = c(x_i, y) - \varrho^i\}. \tag{26}$$

Then $d_\psi < d_0$. Moreover, there exists a dimensional positive constant C such that

$$\mathrm{Lip}(\psi) := \sup_{\substack{y_1, y_2 \in \bar{\Omega}^* \\ y_1 \neq y_2}} \frac{|\psi(y_1) - \psi(y_2)|}{d_{\mathbb{S}^n}(y_1, y_2)} \leq -CF'\left(\frac{d_0 + 3d_\psi}{4}\right) + \frac{8}{d_0 - d_\psi} \max_{\bar{\Omega}^*} |\psi|, \tag{27}$$

where $F \in C^1([0, d_0))$ is given in (6) and (7).

Proof. Given $y \in \bar{\Omega}^*$ and x_i with $\psi(y) = c(x_i, y) - \varrho^i$, we have

$$F(d_{\mathbb{S}^n}(x_i, y)) = c(x_i, y) = \varrho^i + \psi(y_i) \geq \min_{\bar{\Omega}^*} \psi + \min_{1 \leq i \leq N} \varrho^i > -\infty,$$

which implies that $d_\psi < d_0$. To prove (27), let us fix $y_1, y_2 \in \bar{\Omega}^*$ with $y_1 \neq y_2$ and then take $1 \leq k_i \leq N$ satisfying $\psi(y_i) = c(x_{k_i}, y_i) - \varrho^{k_i}$ for each $i = 1, 2$.

Case I. $d_{\mathbb{S}^n}(y_1, y_2) < \frac{1}{4}(d_0 - d_\psi)$.

In this case, $x_{k_1} \in B_{d_0}(y_2)$ and $x_{k_2} \in B_{d_0}(y_1)$, which implies that $d_{\mathbb{S}^n}(x_{k_i}, \gamma(t)) < d_0$ for every $t \in [0, d_{\mathbb{S}^n}(y_1, y_2)]$ and every $i = 1, 2$, where $\gamma(t)$ is the normal geodesic joining y_1 and y_2. Define the dimensional constant

$$C := \sup\left\{\left|\frac{d}{dt} d_{\mathbb{S}^n}(x, \eta(t))\right| : \eta \text{ is a normal geodesic on } \mathbb{S}^n \text{ and } x \in \mathbb{S}^n\right\}. \tag{28}$$

The mean value theorem indicates that there exists $\tau \in (0, d_{\mathbb{S}^n}(y_1, y_2))$ such that

$$\int_0^{d_{\mathbb{S}^n}(y_1, y_2)} \frac{d}{dt} c(x_{k_2}, \gamma(t)) dt = d_{\mathbb{S}^n}(y_1, y_2) \frac{d}{dt}\bigg|_{t=\tau} c(x_{k_2}, \gamma(t)). \tag{29}$$

The triangle inequality gives that

$$\begin{aligned} d_{\mathbb{S}^n}(x_{k_2}, \gamma(\tau)) &\leq d_{\mathbb{S}^n}(x_{k_2}, y_2) + d_{\mathbb{S}^n}(y_2, \gamma(\tau)) \\ &\leq d_{\mathbb{S}^n}(x_{k_2}, y_2) + d_{\mathbb{S}^n}(y_2, y_1) \\ &\leq d_\psi + \frac{1}{4}(d_0 - d_\psi) \end{aligned}$$

$$= \frac{d_0+3d_\psi}{4}. \tag{30}$$

Using (29), (28) and (30) in order, we have

$$\begin{aligned}
&\left|\int_0^{d_{\mathbb{S}^n}(y_1,y_2)} \frac{d}{dt}c(x_{k_2},\gamma(t))dt\right| \\
= \;& d_{\mathbb{S}^n}(y_1,y_2)\left|\frac{d}{dt}\Big|_{t=\tau} c(x_{k_2},\gamma(t))\right| \\
= \;& d_{\mathbb{S}^n}(y_1,y_2)|F'(d_{\mathbb{S}^n}(x_{k_2},\gamma(\tau)))|\left|\frac{d}{dt}\Big|_{t=\tau} d_{\mathbb{S}^n}(x_{k_2},\gamma(t))\right| \\
\leq \;& -CF'\left(\frac{d_0+3d_\psi}{4}\right)d_{\mathbb{S}^n}(y_1,y_2),
\end{aligned}$$

where the concavity of F is used in the inequality. By definition of $\mathcal{A}$, we obtain

$$\begin{aligned}
0 \;=\;& (\psi(y_2)+\varrho^{k_2}-c(x_{k_2},y_2))-(\psi(y_1)+\varrho^{k_2}-c(x_{k_2},y_1)) \\
&+(\psi(y_1)+\varrho^{k_2}-c(x_{k_2},y_1)) \\
\geq\;& \psi(y_2)-\psi(y_1)-c(x_{k_2},y_2)+c(x_{k_2},y_1) \\
=\;& \psi(y_2)-\psi(y_1)-\int_0^{d_{\mathbb{S}^n}(y_1,y_2)}\frac{d}{dt}c(x_{k_2},\gamma(t))dt \\
\geq\;& \psi(y_2)-\psi(y_1)+CF'\left(\frac{d_0+3d_\psi}{4}\right)d_{\mathbb{S}^n}(y_1,y_2).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
0 \;=\;& (\psi(y_2)+\varrho^{k_1}-c(x_{k_1},y_2))-(\psi(y_1)+\varrho^{k_1}-c(x_{k_1},y_1)) \\
&-(\psi(y_2)+\varrho^{k_1}-c(x_{k_1},y_2)) \\
\leq\;& \psi(y_2)-\psi(y_1)+c(x_{k_1},y_1)-c(x_{k_1},y_2) \\
=\;& \psi(y_2)-\psi(y_1)-\int_0^{d_{\mathbb{S}^n}(y_1,y_2)}\frac{d}{dt}c(x_{k_1},\gamma(t))dt \\
\leq\;& \psi(y_2)-\psi(y_1)-CF'\left(\frac{d_0+3d_\psi}{4}\right)d_{\mathbb{S}^n}(y_1,y_2).
\end{aligned}$$

We conclude that

$$\frac{|\psi(y_2)-\psi(y_1)|}{d_{\mathbb{S}^n}(y_1,y_2)} \leq -CF'\left(\frac{d_0+3d_\psi}{4}\right). \tag{31}$$

Case II. $d_{\mathbb{S}^n}(y_1,y_2) \geq \frac{1}{4}(d_0-d_\psi)$.

In this case, we have

$$\frac{|\psi(y_2)-\psi(y_1)|}{d_{\mathbb{S}^n}(y_1,y_2)} \leq \frac{2\max_{\bar\Omega^*}|\psi|}{\frac{1}{4}(d_0-d_\psi)} \leq \frac{8}{d_0-d_\psi}\max_{\bar\Omega^*}|\psi|. \tag{32}$$

Putting (31) and (32) together, we obtain (27). □

To close this section, we show the weak convergence of c-curvature measure with respect to the uniform convergence of c-support functions.

Lemma 2.5. *Consider a sequence $\{(\varrho_j,\psi_j)\}_{j\in\mathbb{N}}$ in $\mathcal{A}$, where ϱ_j converges to a point $\varrho\in\mathbb{R}^N$ and ψ_j uniformly converges to a function $\psi\in C(\bar\Omega^*)$. Then $(\varrho,\psi)\in\mathcal{A}$. Let $\mathbb{K}_j$ and $\mathbb{K}$ be the unbounded convex polyhedra in $\mathscr{K}(X,\mathcal{C})$ associated to (ϱ_j,ψ_j)*

and (ϱ, ψ), respectively. Let $\mu^c_{\mathbb{K}_j}$ and $\mu^c_{\mathbb{K}}$ be the c-curvature measures of $\mathbb{K}_j$ and $\mathbb{K}$, respectively. Then $\mu^c_{\mathbb{K}_j}$ converges to $\mu^c_{\mathbb{K}}$ in the weak sense of measures.

Proof. By Lemma 2.3, it suffices to show that $\lim_{j\to\infty} \mu^c_{\mathbb{K}_j,i} = \mu^c_{\mathbb{K},i}$ for all $1 \le i \le N$, where $\mu^c_{\mathbb{K}_j,i} = |\mathscr{A}^c_{\mathbb{K}_j}(x_i)|$ and $\mu^c_{\mathbb{K},i} = |\mathscr{A}^c_{\mathbb{K}}(x_i)|$. Take $y^j_i \in \mathscr{A}^c_{\mathbb{K}_j}(x_i)$. Suppose $y^j_i \to y_i$ by a subsequence of j. Then $d_{\mathbb{S}^n}(x_i, y_i) < d_0$. Otherwise

$$\psi(y_i) = \lim_{j\to\infty} \psi_j(y^j_i) = \lim_{j\to\infty} \{c(x_i, y^j_i) - \varrho^i_j\} = -\infty,$$

a contradiction. It follows that

$$\varrho^i = \lim_{j\to\infty} \varrho^i_j = \lim_{j\to\infty} \{c(x_i, y^j_i) - \psi_j(y^j_i)\} = c(x_i, y_i) - \psi(y_i),$$

which implies that $y_i \in \mathscr{A}^c_{\mathbb{K}}(x_i)$. Hence $\limsup_{j\to\infty} \mu^c_{\mathbb{K}_j,i} \le \mu^c_{\mathbb{K},i}$.

On the other hand, if $y_i \in \mathscr{A}^c_{\mathbb{K}}(x_i)$, we claim that $y_i \in \mathscr{A}^c_{\mathbb{K}_j}(x_i)$ for any sufficiently large $j \in \mathbb{N}$. Since $y_i \in \mathscr{A}^c_{\mathbb{K}}(x_i)$, we have $\psi(y_i) = \max_{1\le k\le N}\{c(x_k, y_i) - \varrho^k\} = c(x_i, y_i) - \varrho^i$. Combining $\varrho^i = \lim_{j\to\infty} \varrho^i_j$, we see that for any sufficiently large $j \in \mathbb{N}$ there holds

$$\psi_j(y_i) = \max_{1\le k\le N}\{c(x_k, y_i) - \varrho^k_j\} = c(x_i, y_i) - \varrho^i_j,$$

which implies that $y_i \in \mathscr{A}^c_{\mathbb{K}_j}(x_i)$. It follows that $\liminf_{j\to\infty} \mu^c_{\mathbb{K}_j,i} \ge \mu^c_{\mathbb{K},i}$.

Putting two sides together, we complete the proof of Lemma 2.5. □

3. **Oscillation estimates.**

We first give some basic properties of d_0-convex sets. Denote by $\bar{B}_{\pi-d_0}$ a closed geodesic ball in $\mathbb{S}^n$ with radius $\pi - d_0$ and its center unspecified.

Lemma 3.1 (Lemma 2.1 in [10]). *Suppose $\omega \ne \emptyset$ and $\omega \subset \bar{B}_{\pi-d_0}$. Then:*

(i) *$\omega^\sharp$ is d_0-convex and $\omega \subset (\omega^\sharp)^\sharp$;*
(ii) *if ω is d_0-convex, then $\omega = (\omega^\sharp)^\sharp$;*
(iii) *$(\omega^\sharp)^\sharp = \langle\omega\rangle$, where $\langle\omega\rangle$ is the smallest d_0-convex set containing ω;*
(iv) *$\omega^\sharp = \mathbb{S}^n \setminus \omega_{d_0}$, where $\omega_{d_0} = \{x \in \mathbb{S}^n : d_{\mathbb{S}^n}(x, y) < d_0$ for some $y \in \omega\}$.*

We will construct the desired unbounded c-convex polyhedron in $\mathscr{K}$ from the minimizers of functional (25). The technical core for the existence of such minimizers is the oscillation estimates of the minimizing sequence. The following lemma is a strengthened version of condition (ii) of Theorem 1.2, which is an analogue of [3, Lemma 5.11] where the case of $d_0 = \pi/2$ and $\Omega = \Omega^* = \mathbb{S}^n$ was discussed.

Lemma 3.2. *If conditions (i) and (ii) hold, then there exist $\delta \in (0,1)$ and $\alpha \in (0, d_0)$ such that for any compact $\omega \subset \bar{B}_{\pi-d_0}$ with $\omega \cap \{x_1, \dots, x_N\} \ne \emptyset$,*

$$\sum_{x_i\in\omega} \gamma_i < (1-\delta)|\omega_{d_0-\alpha} \cap \Omega^*|. \tag{33}$$

Proof. To prove (33), it suffices to show that there exists a sufficiently small positive number $\delta' \in (0,1)$ such that for any compact $\omega \subset \bar{B}_{\pi-d_0}$ with $\omega \cap \{x_1, \dots, x_N\} \ne \emptyset$,

$$\sum_{x_i\in\omega} \gamma_i < (1-\delta')|\omega_{d_0} \cap \Omega^*|. \tag{34}$$

Suppose (34) holds. Since $\omega \cap \{x_1, \dots, x_N\} \ne \emptyset$, we have

$$|\omega_{d_0} \cap \Omega^*| \ge \min_{1\le i\le N} |B_{d_0}(x_i) \cap \Omega^*| =: \sigma_0 > 0. \tag{35}$$

Fix $0 < \varepsilon < \frac{\sigma_0}{2}\min\{1, \delta'/(1-\delta')\}$ and then let $\delta := \delta' - \varepsilon(1-\delta')\frac{2}{\sigma_0}$. It follows that

$$(1-\delta')\varepsilon = \frac{\sigma_0}{2}(\delta' - \delta). \tag{36}$$

Take a sufficiently small number $\alpha \in (0, d_0)$ such that for any compact $\omega \subset \bar{B}_{\pi-d_0}$ with $\omega \cap \{x_1, \ldots, x_N\} \neq \emptyset$ there holds

$$|\omega_{d_0} \cap \Omega^*| - |\omega_{d_0-\alpha} \cap \Omega^*| < \frac{\sigma_0}{2} < \varepsilon. \tag{37}$$

Applying (36), (37) and (35) in order, we obtain

$$(1-\delta')\varepsilon < (\delta' - \delta)|\omega_{d_0-\alpha} \cap \Omega^*|. \tag{38}$$

Plugging (37) and (38) into (34), we have

$$\begin{aligned}
\sum_{x_i \in \omega} \gamma_i &< (1-\delta')(|\omega_{d_0-\alpha} \cap \Omega^*| + \varepsilon) \\
&= (1-\delta')|\omega_{d_0-\alpha} \cap \Omega^*| + (1-\delta')\varepsilon \\
&< (1-\delta)|\omega_{d_0-\alpha} \cap \Omega^*|.
\end{aligned}$$

Now it suffices to show (34). Suppose on the contrary that there exists a sequence of compact $\omega_j \subset \bar{B}_{\pi-d_0}$ with $\omega_j \cap \{x_1, \ldots, x_N\} \neq \emptyset$ for each $j \in \mathbb{N}$ such that

$$\lim_{j\to\infty} \frac{\sum_{x_i \in \omega_j} \gamma_i}{|(\omega_j)_{d_0} \cap \Omega^*|} = 1. \tag{39}$$

Lemma 3.1 (iii) says that $\langle \omega_j \rangle := \omega_j^{\sharp\sharp}$ is the smallest d_0-convex subset of $\mathbb{S}^n$ containing ω_j. Combining the d_0-convexity of $\omega_j^\sharp$ and Lemma 3.1 (ii), we have

$$(\omega_j)_{d_0} = \mathbb{S}^n \setminus \omega_j^\sharp = \mathbb{S}^n \setminus \langle \omega_j \rangle^\sharp = \langle \omega_j \rangle_{d_0}.$$

It follows that

$$\frac{\sum_{x_i \in \omega_j} \gamma_i}{|(\omega_j)_{d_0} \cap \Omega^*|} \leq \frac{\sum_{x_i \in \langle \omega_j \rangle} \gamma_i}{|\langle \omega_j \rangle_{d_0} \cap \Omega^*|}, \quad \forall j \in \mathbb{N}.$$

Combining condition (ii) and (39), we see that

$$\lim_{j\to\infty} \frac{\sum_{x_i \in \langle \omega_j \rangle} \gamma_i}{|\langle \omega_j \rangle_{d_0} \cap \Omega^*|} = 1.$$

Blaschke's selection theorem yields that by passing to a subsequence, we may assume that the sequence of d_0-convex sets $\langle \omega_j \rangle$ converges to a d_0-convex set ω_∞ with respect to the Hausdorff metric, satisfying

$$\omega_\infty \cap \{x_1, \ldots, x_N\} \neq \emptyset, \quad \sum_{x_i \in \omega_\infty} \gamma_i = |(\omega_\infty)_{d_0} \cap \Omega^*|,$$

which contradicts condition (ii) of Theorem 1.2. $\square$

The following lemma addresses the oscillation estimates of radial functions at vertices for an arbitrary minimizing sequence of $\mathcal{J}$ in the extremal case (12). It is different from the non-extremal case (19) studied in [10].

Lemma 3.3. *Let (ϱ_j, ψ_j) be a minimizing sequence of $\mathcal{J}$ in $\mathcal{A}$. Suppose conditions (i) and (ii) hold. Then there exists a positive constant C independent of j such that*

$$\limsup_{j\to\infty} \left(\max_{1\leq i\leq N} \varrho_j^i - \min_{1\leq i\leq N} \varrho_j^i \right) \leq C, \tag{40}$$

where ϱ_j^i is the i-th component of ϱ_j.

Proof. Recall that $\mathcal{J}$ is invariant under additive constants. Hence we may assume that $\min_{\bar{\Omega}^*} \psi_j = 0$ for each $j \in \mathbb{N}$. Take $y_j \in \bar{\Omega}^*$ with $\psi_j(y_j) = 0$. Let $y_\infty := \lim_{j\to\infty} y_j \in \bar{\Omega}^*$ by passing to a subsequence. Take $y_j^i \in \bar{\Omega}^*$ such that $\varrho_j^i + \psi_j(y_j^i) = c(x_i, y_j^i)$. Then

$$\varrho_j^i = c(x_i, y_j^i) - \psi_j(y_j^i) \le -\min_{\bar{\Omega}^*} \psi_j = 0, \quad \forall 1 \le i \le N. \tag{41}$$

Assume that (40) is not true. Then $s_j := \min_{1\le i\le N} \varrho_i^j \to -\infty$ as $j \to \infty$ along a subsequence. Without loss of generality, we may assume that $s_j = \varrho_j^1$ for each $j \in \mathbb{N}$.

Step 1. Estimate $\int_{\Omega^*} \psi_j(y) d\sigma_{\mathbb{S}^n}(y)$.

Using $\psi_j(y) \ge c(x_1, y) - \varrho_j^1$ for each point $y \in \Omega^* \cap B_{d_0}(x_1)$ and $\psi_j \ge 0$, we have

$$\begin{aligned} \int_{\Omega^*} \psi_j(y) d\sigma_{\mathbb{S}^n}(y) &\ge \int_{\Omega^* \cap B_{d_0}(x_1)} \{c(x_1, y) - \varrho_j^1\} d\sigma_{\mathbb{S}^n}(y) \\ &\ge -\sigma s_j - C, \end{aligned} \tag{42}$$

where $\sigma := |\Omega^* \cap B_{d_0}(x_1)| > 0$ and C is a positive constant independent of j.

Step 2: Estimate $\sum_{x_i \in B_{d_0}(y_\infty)} \gamma_i \varrho_j^i$.

Note that (18) yields $B_{d_0}(y_\infty) \cap X \neq \emptyset$. Without loss of generality, we may assume $x_i \in B_{d_0}(y_\infty)$ if and only if $1 \le i \le N_1$ with $N_1 < N$. Using $y_\infty = \lim_{j\to\infty} y_j$, we see that $B_{d_0}(y_j)$ converges to $B_{d_0}(y_\infty)$ with respect to the Hausdorff metric. In particular, we have $x_1, \ldots, x_{N_1} \in B_{d_0}(y_j)$ for sufficiently large $j \in \mathbb{N}$. Then

$$\varrho_j^i \ge c(x_i, y_j) - \psi(y_j) = c(x_i, y_j), \quad \forall 1 \le i \le N_1.$$

Let $\varepsilon := d_0 - \max_{1\le i\le N_1} d_{\mathbb{S}^n}(x_i, y_\infty) > 0$. Then for sufficiently large $j \in \mathbb{N}$ there holds

$$c(x_i, y_j) \ge F\left(d_0 - \frac{\varepsilon}{2}\right), \quad \forall 1 \le i \le N_1.$$

It follows that

$$\sum_{x_i \in B_{d_0}(y_\infty)} \gamma_i \varrho_j^i = \sum_{i=1}^{N_1} \gamma_i \varrho_j^i \ge F\left(d_0 - \frac{\varepsilon}{2}\right) \sum_{i=1}^{N_1} \gamma_i \ge -C. \tag{43}$$

Step 3: Estimate $\sum_{x_i \in V_{y_\infty, d_0}} \gamma_i \varrho_j^i$, where $V_{y_\infty, d_0} := \mathbb{S}^n \setminus B_{d_0}(y_\infty)$. We aim to show that there exist constants $\delta \in (0, 1)$ and $C > 0$ independent of j such that

$$\sum_{x_i \in V_{y_\infty, d_0}} \gamma_i \varrho_j^i > (\delta - 1) \int_{\Omega^*} \psi_j(y) d\sigma_{\mathbb{S}^n}(y) - C. \tag{44}$$

It follows from Step 2 that $x_i \in V_{y_\infty, d_0}$ if and only if $N_1 + 1 \le i \le N$. By reordering the index, we may assume that

$$\varrho_j^{N_1+1} \le \varrho_j^{N_1+2} \le \cdots \le \varrho_j^N \le 0, \tag{45}$$

where (41) is used in the last inequality. Let $\delta \in (0, 1)$ and $\alpha \in (0, d_0)$ be the constants given in Lemma 3.2. Set $W_{N_1+1} := B_{d_0-\alpha}(x_{N_1+1})$ and

$$W_k := \Big(\bigcup_{i=N_1+1}^{k} B_{d_0-\alpha}(x_i) \Big) \setminus \Big(\bigcup_{i=N_1+1}^{k-1} B_{d_0-\alpha}(x_i) \Big), \quad \forall N_1 + 2 \le k \le N.$$

Lemma 3.2 yields that for any $N_1+1 \le k \le N$,

$$\sum_{i=N_1+1}^{k} \gamma_i < (1-\delta)\Big|\Big(\bigcup_{i=N_1+1}^{k} B_{d_0-\alpha}(x_i)\Big)\cap\Omega^*\Big| = (1-\delta)\sum_{i=N_1+1}^{k} |W_i\cap\Omega^*|. \quad (46)$$

Using (45) and (46), we have

$$\begin{aligned}
&\sum_{x_i\in V_{y_\infty,d_0}} \gamma_i\varrho_j^i \\
= \;& \sum_{i=N_1+1}^{N} \gamma_i\varrho_j^i = \varrho_j^N \sum_{i=N_1+1}^{N}\gamma_i + \sum_{k=N_1+1}^{N-1}\Big((\varrho_j^k-\varrho_j^{k+1})\sum_{i=N_1+1}^{k}\gamma_i\Big) \\
> \;& (1-\delta)\Big[\varrho_j^N \sum_{i=N_1+1}^{N}|W_i\cap\Omega^*| + \sum_{k=N_1+1}^{N-1}\Big((\varrho_j^k-\varrho_j^{k+1})\sum_{i=N_1+1}^{k}|W_i\cap\Omega^*|\Big)\Big] \\
= \;& (1-\delta)\sum_{i=N_1+1}^{N}\varrho_j^i|W_i\cap\Omega^*|. \qquad (47)
\end{aligned}$$

Take $\breve{y}_j^i \in B_{d_0-\alpha}(x_i)$ such that $\psi_j(\breve{y}_j^i) = \min_{\bar{B}_{d_0-\alpha}(x_i)\cap\bar{\Omega}^*}\psi_j$. We have

$$\varrho_j^i \ge c(x_i,\breve{y}_j^i) - \psi_j(\breve{y}_j^i) \ge\; F(d_0-\alpha) - \psi_j(\breve{y}_j^i). \quad (48)$$

Plugging (48) into (47), we have

$$\begin{aligned}
\sum_{x_i\in V_{y_\infty,d_0}} \gamma_i\varrho_j^i \;&>\; (\delta-1)\int_{\cup_{i=N_1+1}^{N}W_i\cap\Omega^*}\psi_j(\breve{y}_j^i)d\sigma_{\mathbb{S}^n}(y) - C \\
&\ge\; (\delta-1)\int_{\cup_{i=N_1+1}^{N}W_i\cap\Omega^*}\psi_j(y)d\sigma_{\mathbb{S}^n}(y) - C.
\end{aligned}$$

Combining $\min_{\bar{\Omega}^*}\psi_j = 0$, we obtain (44).

Step 4: Show that $\mathcal{J}(\varrho_j,\psi_j)\to\infty$ as $j\to\infty$.

Putting (42), (43) and (44) together, we see that

$$\begin{aligned}
\mathcal{J}(\varrho_j,\psi_j) \;&=\; \int_{\Omega^*}\psi_j(y)d\sigma_{\mathbb{S}^n}(y) + \sum_{x_i\in B_{d_0}(y_\infty)}\gamma_i\varrho_j^i + \sum_{x_i\in V_{y_\infty,d_0}}\gamma_i\varrho_j^i \\
&\ge\; \int_{\Omega^*}\psi_j(y)d\sigma_{\mathbb{S}^n}(y) - C + (\delta-1)\int_{\Omega^*}\psi_j(y)d\sigma_{\mathbb{S}^n}(y) \\
&\ge\; -\delta\sigma s_j - C \to \infty, \quad \text{as } j\to\infty.
\end{aligned}$$

This contradicts that (ϱ_j,ψ_j) is a minimizing sequence of $\mathcal{J}$ in $\mathcal{A}$. We complete the proof. ∎

Lemma 3.4. *Suppose conditions (i) and (ii) hold. Let (ϱ_j,ψ_j) be a minimizing sequence of $\mathcal{J}$ in $\mathcal{A}$. Then there exists a positive constant C independent of j such that*

$$\limsup_{j\to\infty} \operatorname{osc}_{\bar{\Omega}^*}\psi_j \le C. \quad (49)$$

Proof. Recall that $\mathcal{J}$ is invariant under additive constants. Hence we may assume that $\min_{1\le i\le N}\varrho_j^i = 0$ for each $j\in\mathbb{N}$. Then Lemma 3.3 yields $\max_{1\le i\le N}\varrho_j^i \le C$ for each $j\in\mathbb{N}$. Take $\hat{y}_j\in\bar{\Omega}^*$ with $\psi_j(\hat{y}_j) = \max_{\bar{\Omega}^*}\psi_j$. Using (21), we see that there exists $1 \leq k_j \leq N$ such that $\psi_j(\hat{y}_j) = c(x_{k_j}, \hat{y}_j) - \varrho_j^{k_j}$. Combining the concavity of F,

$$\max_{\bar{\Omega}^*} \psi_j \leq -\min_{1\leq i\leq N} \varrho_j^i \leq 0.$$

We will prove (49) using the contradiction argument. Assume on the contrary that $\lim_{j\to\infty} \operatorname{osc}_{\bar{\Omega}^*} \psi_j = \infty$. Then $s_j := \min_{\bar{\Omega}^*} \psi_j \to -\infty$ as $j \to \infty$. Take $y_j \in \bar{\Omega}^*$ such that $s_j = \psi_j(y_j)$. By passing to a subsequence we may set $y_\infty := \lim_{j\to\infty} y_j \in \bar{\Omega}^*$. Note that (18) yields $B_{d_0}(y_\infty) \cap X \neq \emptyset$. Without loss of generality, we may assume that $x_i \in B_{d_0}(y_\infty)$ if and only if $1 \leq i \leq N_1$ with $1 \leq N_1 < N$. Since $0 \leq \varrho_j^i \leq C$, we see that $\sum_{i=1}^{N_1} \gamma_i \varrho_j^i$ is uniformly bounded for each $j \in \mathbb{N}$.

On the other hand, note that $B_{d_0}(y_j)$ converges to $B_{d_0}(y_\infty)$ in the Hausdorff metric. Then for any sufficiently large $j \in \mathbb{N}$ there holds $x_i \in B_{d_0}(y_j)$ for each $1 \leq i \leq N_1$. Let $\varepsilon := d_0 - \max_{1\leq i\leq N_1} d_{\mathbb{S}^n}(x_i, y_\infty) > 0$. Then for any sufficiently large $j \in \mathbb{N}$,

$$c(x_i, y_j) \geq F\left(d_0 - \frac{\varepsilon}{2}\right), \quad \forall 1 \leq i \leq N_1.$$

Using (21), we have $\varrho_j^i \geq c(x_i, y_j) - s_j$ for each $1 \leq i \leq N_1$. It follows that

$$\begin{aligned}
\sum_{i=1}^{N_1} \gamma_i \varrho_j^i &\geq \sum_{i=1}^{N_1} \gamma_i c(x_i, y_j) - s_j \sum_{i=1}^{N_1} \gamma_i \\
&\geq -s_j \sum_{i=1}^{N_1} \gamma_i + F\left(d_0 - \frac{\varepsilon}{2}\right) \sum_{i=1}^{N_1} \gamma_i \to \infty, \quad \text{as } j \to \infty,
\end{aligned}$$

which contradicts that $\sum_{i=1}^{N_1} \gamma_i \varrho_j^i$ is uniformly upper bounded for each $j \in \mathbb{N}$. $\square$

4. **Proof of Theorem 1.2.** Let (ϱ_j, ψ_j) be a minimizing sequence of $\mathcal{J}$ in $\mathcal{A}$. Since $\mathcal{J}$ is invariant under additive constants, we may safely assume that $\min_{\bar{\Omega}^*} \psi_j = 0$ for each $j \in \mathbb{N}$. Note that (41) implies $\max_{1\leq i\leq N} \varrho_j^i \leq 0$. Then Lemma 3.3 indicates $\min_{1\leq i\leq N} \varrho_j^i$ is uniformly bounded from below. Hence there exists a positive constant C independent of j such that

$$\limsup_{j\to\infty} |\varrho_j| = \limsup_{j\to\infty} \sqrt{\sum_{i=1}^N |\varrho_j^i|^2} \leq C. \tag{50}$$

On the other hand, Lemma 3.4 shows that $\max_{\bar{\Omega}^*} \psi_j = \operatorname{osc}_{\bar{\Omega}^*} \psi_j$ is uniformly upper bounded. Then there exists a positive constant C independent of j such that

$$\limsup_{j\to\infty} \max_{\bar{\Omega}^*} |\psi_j| \leq C. \tag{51}$$

Given any $y \in \bar{\Omega}^*$, there exists $1 \leq i_j \leq N$ such that $\psi_j(y) = c(x_{i_j}, y) - \varrho_j^{i_j}$. Then

$$c(x_{i_j}, y) = \psi_j(y) + \varrho_j^{i_j} \geq \min_{\bar{\Omega}^*} \psi_j + \min_{1\leq i\leq N} \varrho_j^i \geq -C.$$

Since $y \in \bar{\Omega}^*$ is arbitrary, we see that there exists a sufficiently small positive number ε independent of j such that $d_{\psi_j} \leq d_0 - \varepsilon$ for each $j \in \mathbb{N}$, where d_{ψ_j} is defined as in (26). Lemma 2.4 indicates there exists a positive constant C independent of j such that

$$\limsup_{j\to\infty} \operatorname{Lip}(\psi_j) \leq C. \tag{52}$$

Putting (50), (51) and (52) together, and using the Arzelá-Ascoli Lemma, we see that (ϱ_j, ψ_j) has a convergent subsequence, which will still be denoted by (ϱ_j, ψ_j). Define $\psi_\infty := \lim_{j\to\infty} \psi_j$ and $\varrho_\infty := \lim_{j\to\infty} \varrho_j$. Then $(\varrho_\infty, \psi_\infty)$ minimizes $\mathcal{J}$ in $\mathcal{A}$. Lemma 2.2 says there is an unbounded c-convex polyhedron $\mathbb{K}_\infty \in \mathscr{K}$ associated to $(\varrho_\infty, \psi_\infty)$.

It suffices to show that $\mathbb{K}_\infty$ satisfies $\mu^c_{\mathbb{K}_\infty} = \gamma$, where $\mu^c_{\mathbb{K}_\infty}$ is the c-curvature measure of $\mathbb{K}_\infty$. By virtue of Lemma 2.3, this is equivalent to $\mu^c_{\mathbb{K}_\infty,i} := |\mathscr{A}^c_{\mathbb{K}_\infty}(x_i)| = \gamma_i$ for each $1 \le i \le N$. Given a small $\varepsilon > 0$, define $\varrho^1_\varepsilon := \varrho_\infty + \varepsilon$ and $\varrho^i_\varepsilon := \varrho^i_\infty$ for each $2 \le i \le N$. We also define $\psi_\varepsilon(y) := \max_{1\le i\le N}\{c(x_i, y) - \varrho^i_\varepsilon\}$ for each $y \in \bar{\Omega}^*$. Using Lemma 2.2 again, the pair $(\varrho_\varepsilon, \psi_\varepsilon) \in \mathcal{A}$ determines an unbounded c-convex polyhedron $\mathbb{K}_\varepsilon \in \mathscr{K}(X, \mathcal{C})$ such that $\varrho_\varepsilon = -\log r_{\mathbb{K}_\varepsilon}(y_i)$ and $\psi_\varepsilon = \log u_{\mathbb{K}_\varepsilon}$. It is clear that $\mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_1) \subset \mathscr{A}^c_{\mathbb{K}_\infty}(x_1)$ and $\mathscr{A}^c_{\mathbb{K}_\infty}(x_i) \subset \mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_i)$ for each $2 \le i \le N$. We also have

$$
\begin{aligned}
\psi_\varepsilon(y) &= \psi_\infty(y), \quad \text{if } y \in \mathscr{A}^c_{\mathbb{K}_\infty}(x_i) \text{ with } 2 \le i \le N;\\
\psi_\varepsilon(y) &\le \psi_\infty(y), \quad \text{if } y \in \mathscr{A}^c_{\mathbb{K}_\infty}(x_1) \setminus \mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_1);\\
\psi_\varepsilon(y) &= c(x_1, y) - \varrho^1_\varepsilon \le \psi_\infty(y) - \varepsilon, \quad \text{if } y \in \mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_1).
\end{aligned}
$$

Combining the fact that $(\varrho_\infty, \psi_\infty)$ minimizes $\mathcal{J}$ in $\mathcal{A}$, we have

$$
\begin{aligned}
0 &\le \frac{1}{\varepsilon}(\mathcal{J}(\psi_\varepsilon, \varrho_\varepsilon) - \mathcal{J}(\psi_\infty, \varrho_\infty))\\
&= \gamma_1 + \frac{1}{\varepsilon}\Big(\int_{\mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_1)} + \int_{\mathscr{A}^c_{\mathbb{K}_\infty}(x_1)\setminus\mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_1)} + \sum_{2\le i\le N}\int_{\mathscr{A}^c_{\mathbb{K}_\infty}(x_i)}\Big)(\psi_\varepsilon - \psi_\infty)d\sigma_{\mathbb{S}^n}\\
&\le \gamma_1 - \mu^c_{\mathbb{K}_\varepsilon,1},
\end{aligned}
$$

where $\mu^c_{\mathbb{K}_\varepsilon,1} := |\mathscr{A}^c_{\mathbb{K}_\varepsilon}(x_1)|$. Sending $\varepsilon \to 0$ and using Lemma 2.5, we further obtain

$$
\mu^c_{\mathbb{K}_\infty,1} \le \gamma_1.
$$

The inequalities for $2 \le i \le N$ are derived similarly. Using $\sum_{i=1}^N \mu^c_{\mathbb{K}_\infty,i} = \sum_{i=1}^N \gamma_i$, we have $\mu^c_{\mathbb{K}_\infty,i} = \gamma_i$ for each $1 \le i \le N$. This completes the proof of Theorem 1.2.

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